

HYEONMIN JEON

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Applying to Ph.D. programs in IE / OR, Fall 2027

EDUCATION

Yonsei University, Department of Industrial Engineering Seoul, Korea
M.S. in Industrial Engineering — Financial Investment AI Lab, advisor Prof. Kyong Joo Oh Mar 2025 – Feb 2027 (expected)

- GPA 3.97 / 4.0 (letter-grade basis)
- Relevant coursework (all A range): Stochastic Modelling Theory, Probabilistic Models & Bayesian Inference, Time-Series Analysis, Deep Learning Theory & Applications, Advanced Investment Engineering, Financial Engineering Topics, Corporate Finance

Chung-Ang University, School of Electrical and Electronics Engineering Seoul, Korea
B.S. in Electrical and Electronics Engineering Mar 2019 – Feb 2025

- GPA 3.59 / 4.0 (letter-grade basis) · Graduation thesis: *Light-weighting Tiny YOLOv3: The Impact of Quantization on Real-World Applications* (advisor Prof. Minhyeok Lee)
- Relevant coursework (all A or A+): Linear Algebra, Mathematics for Electrical Engineering, Numerical Analysis, Random Variables, Signals & Systems, Machine Learning, Artificial Intelligence, Algorithms

RESEARCH INTERESTS

Bayesian learning and simulation optimization under input uncertainty; learned and multi-fidelity surrogate models; generative models (diffusion, normalizing flows) as simulation input models; sequential decision-making with statistical guarantees.

RESEARCH EXPERIENCE

Financial Investment AI Lab, Yonsei University Seoul, Korea
Graduate Research Assistant Mar 2025 – Present

- Generative scenario models for portfolio risk: a conditional diffusion transformer that generates joint multi-asset return scenarios, evaluated by the risk decisions (VaR/ES reserves) made on its scenarios rather than by sample realism (first-author manuscript, under review at *Expert Systems with Applications*).
- Timing in the equity cross-section: predicting complex wavelet scalograms so that the phase of price moves, and hence when each stock reprices, is preserved and used for cross-sectional ranking (first-author manuscript, under review at *Expert Systems with Applications*).
- Learned priors for Bayesian factor selection: replacing the mixture-of-g prior with a normalizing flow that keeps an explicit density, checked against the exact posterior over models (TV distance 0.016), then tuned by Bayesian optimization; compared with a density-free GAN prior (working paper).
- Ranking and selection under input uncertainty: implemented and verified procedures (KN, feasibility determination, recycled observations, CRN) and studied how a fitted input model affects the probability-of-correct-selection guarantee.
- Co-authored lab manuscripts on deep convolutional factor models with explainable AI (2nd author) and RL-based enhanced indexing (3rd author).

TIPS R&D Project with Urban Complex (lab project)

Seoul, Korea

Researcher — artist-style generative engine

Mar 2025 – Aug 2026

- Compared three vision-language captioners for artistic style and built a structured captioning pipeline; trained a QLoRA style classifier on Qwen2-VL-7B (91.4 % seen / 76.0 % unseen; 96.0 % fused with a DINOv2 prior).
- Moved StyleCrafter generation from VideoCrafter to SDXL, added domain checkpoints routed by the classifier's verdict, and specified the service API, evaluation dashboard and 3D (STL) output path. With J. Jeon and D. H. You.

GAIL (AI Lab), School of EEE, Chung-Ang University

Seoul, Korea

Undergraduate Researcher

Dec 2022 – Dec 2024

- Diffusion-model oversampling for imbalanced fraud data (KICS Fall Conference 2023); latent-diffusion portfolio study; lifelog sleep-score prediction (ETRI challenge); 3D point-cloud classification.

Carbon-Neutral ESG Practical-Problem Research Group, Chung-Ang University

Seoul, Korea

Team Lead, "Green-Navi" (Green AI track)

Apr 2023 – Dec 2023

- Reinforcement learning for fuel-optimal urban routing in AIMSUN; presented at the research fair (Nov 2023). MSIT / NRF Field-Linked Future-Leading Talent programme.

PUBLICATIONS & PRESENTATIONS

SUBMITTED TO JOURNALS

1. **H. M. Jeon**, H. S. Lee*, K. J. Oh*. Conditional Diffusion Transformers for Multi-Asset Scenario Generation and Portfolio Risk Measurement. *Expert Systems with Applications* · ESWA-D-26-39029. Under review, 2026.
2. **H. M. Jeon**, H. S. Lee*, K. J. Oh*. FinPhasor: Phase-Preserving Measurement of Cross-Sectional Timing in Equity Markets. *Expert Systems with Applications* · ESWA-D-26-35395. Under review, 2026.
3. S. H. Kim, **H. M. Jeon**, D. H. You, J. Jeon, S. Baek, H. S. Lee, K. J. Oh*. Deep Convolutional Factor Prediction with Explainable AI for Enhanced Investment Strategy. *Applied Soft Computing* · previously *Expert Systems with Applications* · next venue pending. Pending resubmission, 2026.09.
4. D. H. You, J. Jeon, **H. M. Jeon**, S. H. Kim, S. H. Yoo, T. Y. Kim, H. S. Lee, K. J. Oh*. Benchmark-Aware Enhanced Indexing Guided by the Volume–Price Index. *JCIS* · previously *Expert Systems with Applications* (ESWA-D-25-33272) · next venue pending. Pending resubmission, 2025.12.

CONFERENCE PRESENTATIONS AND COMPETITION SUBMISSIONS

5. Y. Jeong*, H. Kim, D. H. You, **H. M. Jeon**, A. de Castro, T. Lee. A Study for Forecasting Solar Energy Power Generation. *45th International Symposium on Forecasting (ISF 2025)*, Beijing, Jul 2025. Talk; with SAS Institute.
6. **H. M. Jeon**, M. Lee. Dealing with the Imbalanced-Data Problem with a Diffusion Model. *KICS Fall Conference*, Nov 2023.
7. **H. M. Jeon** et al. Personalized Sleep Score Prediction Model Using a Lifelog Dataset. Submitted to the *ETRI Human Understanding AI Paper Challenge*, 2024.

ONGOING RESEARCH

8. **H. M. Jeon**. Learned Surrogate Priors for Bayesian Model Selection: Density, Uncertainty, and the Cost of a Sampler. Working paper; lab seminar, Jan 2026.
9. **H. M. Jeon**. Bayesian Capacity Sizing for an Exchange Matching Queue under Over-Dispersed Arrivals. Working paper, 2026.

TEACHING

- Graduate Program in AI Investment Engineering** (AI인베스트먼트협동과정, contract-based graduate program), Yonsei University Seoul, Korea
Teaching Assistant Sep 2025 – Present
- Financial Engineering** (undergraduate), Department of Industrial Engineering, Yonsei University Seoul, Korea
Teaching Assistant Spring 2026

SELECTED PROJECTS

- Retina Exam Room Scheduling**, with Nune Eye Hospital Feb 2026 – Present
- Recalibrated a discrete-event simulator to the hospital's records; rule, behaviour-cloned and PPO policies on a 44-day fair replay (time in room 12.2 → 8.2 min); read-only EMR integration and live comparison (wait between exams 2.73 → 1.14 min).
- Lucubra** — research-agent platform Sep 2026 – Present
- LLM agents that verify papers, replicate numbers and draft referee reports against a journal baseline; referee committees, review ledger, MCP server (Next.js, SQLite, Anthropic SDK).
- Simulation-methodology reimplementations** — rs-lab, Input-Uncertainty Select, TailFlow, roughvol-lab, Eco-Route RL, Splat from Scratch Sep 2026
- Ranking & selection with verified guarantees; input-uncertainty effects on selection; conditional diffusion as a risk engine; rough volatility; CO₂-optimal routing on SUMO with P(correct selection) 0.956; 3D Gaussian splatting in plain PyTorch.
- Light-weighting Tiny YOLOv3** — graduation thesis, Chung-Ang University 2024
- Int8 quantization with layer fusion; total footprint 1.6 GB → 100 MB (weight file 4× smaller), inference 22–64 % faster across five devices, mAP –2.78 pt on PASCAL VOC. Team of three.
- fai.yonsei.ac.kr** — the lab's website Jul 2025
- Built and maintain the Financial Investment AI Lab's website (React, Vite).

HONORS & AWARDS

- Special Prize (data use)**, Chungcheong ICT Innovation public-data competition — GIS and K-water data on regional waste management 2024
- DB Financial Economics paper competition** — a national-pension financial-product proposal (with H. Kim) 2024

SKILLS

- PROGRAMMING** Python (NumPy, SciPy, pandas, PyTorch, CUDA) | C++ | MATLAB | TypeScript / Next.js | SQL | LaTeX | Linux multi-GPU servers
- METHODS** Discrete-event & Monte Carlo simulation | Ranking & selection | Variance reduction | Diffusion & flow models | Reinforcement learning | Behaviour cloning | Bayesian inference | Time-series analysis
- TOOLS** Eclipse SUMO | AIMSUN | SAS | Hugging Face | Electron | PostgreSQL / SQLite
- LANGUAGES** Korean (native) | English (working proficiency; TOEFL in preparation)